Accessibility Through AI

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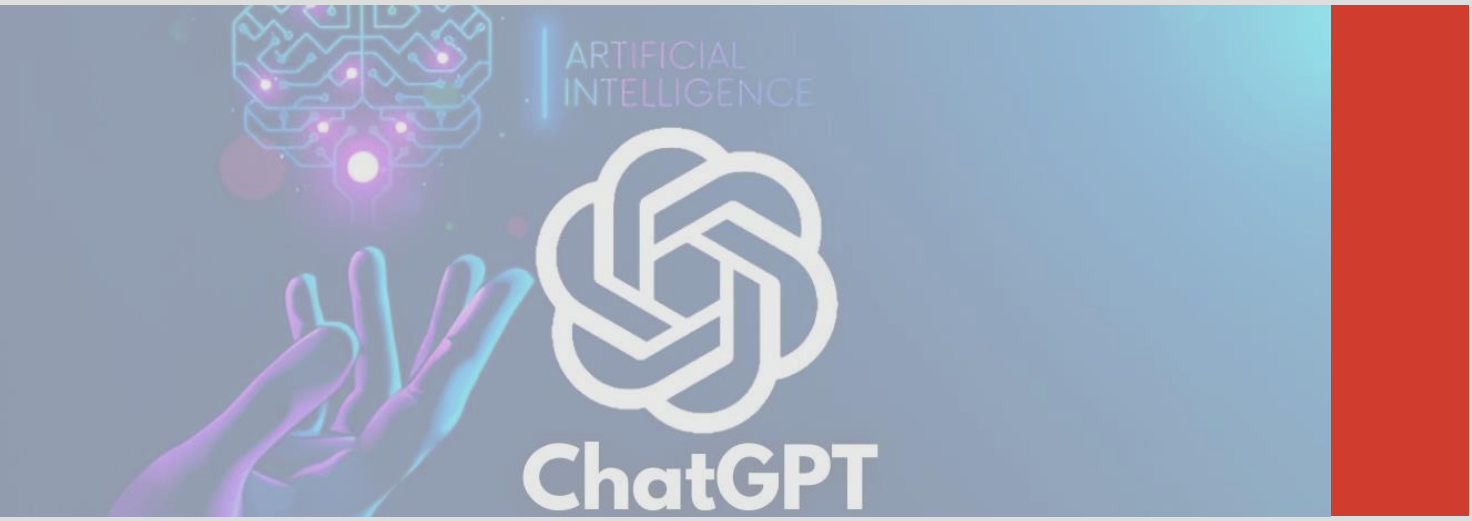

Canadian Urban Institute
Institut Urbain du Canada

Rick Hansen Foundation

Introduction

As our urban environments have expanded rapidly, so has our understanding of accessible space and design. With this growth comes an increasing need for stakeholders within Toronto, as well as throughout the world's modern cities to analyze and evaluate accessibility. With huge portions of our population experiencing physical or mental impairments, the need for inclusionary design has become ever present. To facilitate this, cities like Toronto manually check for accessibility features in the public realm, alongside public consultation to evaluate accessibility in the city.

This report explores the possibility of using artificial intelligence (specifically ChatGPTs image analysis model) to analyse Google Street View (GSV) images in order to catalog and evaluate accessibility features. By supplying AI with 100s of pictures sourced from GSV throughout Toronto, I iterated on a prompt which laid out the criteria for evaluation, as well as the physical features along the street I was looking for. My goal is to provide a use-case for AI in accessibility evaluation in a city such as Toronto.

The Canadian Urban Institute and the Rick Hansen Foundation provided advice, mentorship and guidance. Their direction pushed me towards a better understanding of accessibility and the intersection with built environments for which I was truly grateful.

Interim Summary

Testing ChatGPT's capabilities using Google Street View images focused on its ability to interpret legislation, analyze visuals, and provide actionable recommendations. This involved assessing ChatGPT's ability to analyze accessibility legislation such as AODA, TADG, CSA, and RHF, ensuring alignment with established standards and applicability to my prompts. Image examination tested ChatGPT capacity to identify barriers such as uneven sidewalks, missing curb cuts, or narrow pathways and assess their impact on accessibility. This proposed practical solutions, such as installing tactile paving or accessible ramps, designed to comply with relevant guidelines. Prompts included copying text directly from these guidelines to test comprehension solutions for visualized barriers, to ensure that ChatPT is able to connect legislation to accessibility issues.

The results showed varied responses, with initial inconsistencies in interpreting precise measurements such as width and height to images, often due to a lack of context or poor image quality. To improve this, more precise prompts were developed to provide better accuracy and minimize vagueness. I introduced a scoring framework to evaluate ChatGPT's performance, focusing on the presence and quality of identified features through actionable recommendations. These refinements led to significant improvements in ChatGPT's evaluations, making responses more consistent and accurate. However, ChatGPT's effectiveness was still influenced by image quality, with poor visuals sometimes limiting how precise it was.

Ontario
ACCESSIBILITY FOR ONTARIANS WITH DISABILITIES ACT (AODA)

TORONTO

CSA GROUP

Rick Hansen Foundation

Interim Summary

Through the weeks of testing ChatGPT capabilities, I was able to identify strengths and weaknesses of this technology. ChatGPT was able to correctly identify many physical accessibility features on the images, such as the presence of street furniture, crosswalks, ramps and others. It was also able to provide general insights by analyzing the images and improved its response accuracy through refined and repeated prompts (see Figure 1). This means that ChatGPT can learn and improve through testing more images and data. When ChatGPT identified an absence of a specific accessibility feature, it provided effective suggestions for its improvement. However, ChatGPT also has many limitations. This technology was not able to process large documents and extract needed information from them. The generic responses lacked depth and were not helpful in identifying the quality of specific features. Some of ChatGPT responses lacked accuracy due to obstruction in the images. In addition, ChatGPT was not able to measure specific features such as the width of crosswalks and heights of objects.

Figure 1: Testing for Accuracy

PROMPT:
How many entrances have steps?

RESPONSE:
From the image provided, two entrances clearly have steps:

1. The entrance to the "Bergström" store on the far right has a visible step.
2. The entrance next to it, with the "For Lease" sign, also has a step.

Other entrances, like the ones for RE/MAX, appear to be level with the sidewalk and may not have steps. Therefore, two entrances are not accessible based on the presence of steps.

Prompt Engineering

The purpose of initial prompts was to understand how ChatGPT could evaluate the presence and quality of features. I tested ChatGPT's capabilities in a variety of areas, including legislation comprehension, image analysis, and generating recommendations. The results of this were varied, with some tasks performing better than others. Early observations indicated that the measuring accuracy and processing documentation could pose challenges moving forward. This highlighted potential limitations that would need further attention in the testing process. My first prompt used information from AODA accessibility standards and asked ChatGPT to access the image based on that information. My first prompt was:

"Does the intersection in the image follow these AODA standards "Minimum clear width The minimum clear width of a new or redeveloped outdoor sidewalk or walkway must be 1,500 mm. This is wide enough to accommodate a greater range of two way traffic, including pedestrians who use mobility devices. The minimum 1,500 mm clear width must be free from any obstructions. Any obstructions such as advertising boards, planters and newspaper boxes must be placed outside of the pedestrian route to meet the minimum 1,500 mm clear width requirement. Where there is a curb ramp on the sidewalk or walkway, the minimum clear width can be reduced to 1,200 mm to accommodate the flared sides and base ramp features of the curb ramp design."

I worked on refining my approach by developing precise prompts that minimize unnecessary inputs or comments from ChatGPT. One of the key strategies that I implemented was providing a clear, structured list of requirements, asking ChatGBT to evaluate each aspect individually and assign one point for every criterion made. Once the individual parts were given, I asked ChatGPT to calculate a percentage score based on the total points earned. This process helped me make the assessment more reliable and standardize the evaluation process. By being more specific with my prompts, I was able to enhance the accuracy of ChatGPT's responses when evaluating if images met accessibility standards derived from guidelines such as AODA, TADG, and RHF. The prompt also scored each feature for their presence, quality, and absence from 0 to 1. After conconducting the accuracy evaluations, the final prompt was adjusted to account for grouped criteria and a score was added for features that were obstructed or unclear in the image. A score of 0 was given if the feature did not meet the accessibility standards, 1 was given if the standards were met, N/A was given if the feature was not present and 99 was given if the feature could not be identified due to obstruction in the image. The structure of the final prompt consisted of the initial instruction followed by a point structure. After, each feature was coded for geoJSON and then the criteria for each feature.

Final Prompt

Analyze the photo using the criteria below. Give 1 point for each criteria that is met. Give 0 points for each criteria that is not met. If a feature is not present, give all the corresponding criteria a score of N/A and give a score of 99 if the feature is present but there are obstructions or the image quality is so poor that it prevents determining the criterion's condition. At the end, sort results in json.

json = {key 1: value 1, key 2: value 2}

Sdwk_width: The sidewalk or path should be wide enough to accommodate mobility devices. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if no sidewalks are present.
Sdwk_2_dev: Sidewalk is wide enough for two mobility devices. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if no sidewalks are present.
Sdwk_stable: The surface of the sidewalk must be stable with a slope of no more than 1:20. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if no sidewalks are present.
Sdwk_firm: The surface of the sidewalk must be firm with a slope of no more than 1:20. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if no sidewalks are present.
*Sdwk_drain**: The sidewalk should have adequate drainage to prevent water pooling or erosion. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if no sidewalks are present.
TWSI_curbc: TWSI are present on curb cuts leading into traffic. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if no curb cuts are present.
TWSI_con: provide visual contrast with the surrounding surface. TWSI can be yellow or dark brown. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if no TWSI are present.
TWSI_bumps: TWSI have raised bumps to provide tactile feedback for pedestrians with visual impairments. Note that TWSI may be yellow or red or brown. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if no TWSI are present.
TWSI_dir: TWSI should guide pedestrians accurately across crossings or changes in direction. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if no TWSI are present.
Cross_curb: The pedestrian crosswalk should have smooth curb cuts. Give a score of 1 if there is a smooth curb ramp, 0 if there are no curb ramps, and N/A if it should not belong there because it is not a crosswalk.
Cross_con:Pedestrian crosswalks should be clearly marked with high-contrast colours. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if no pedestrian crosswalks are present.
Cross_pshb Accessible pedestrian push buttons are present. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if no pedestrian crosswalks are present.
Trnst_brdg: Transit stops must feature an unobstructed ground area with enough space for mobility device maneuverability. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if no transit stops are present.
Trnst_wea: Designated waiting transit shelters should have a roof or overhang. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if no transit shelter is present.
Trnst_rain:Designated waiting transit shelters should at minimum have a roof or overhang to protect from rain. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if no transit shelter is present.

Final Prompt

Trnst_wind:Designated waiting transit shelters have glass walls. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if no transit shelter is present.
Trnst_seat:Designated waiting transit shelters should provide accessible seating. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if no transit shelter is present.
Trnst_spc:Designated waiting transit shelters should provide enough space for mobility devices. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if no transit shelter is present.
Trnst_vis: Transit shelters must allow for clear visibility of approaching vehicles (e.g., clear walls, proper positioning). Give a score of 1 if satisfied, 0 if not satisfied, and N/A if no transit shelter is present.
Trnst_sign: Signage with clear, visible, and accessible information (e.g., route numbers, schedules) must be present. Give a score of 1 if there is clear visible and accessible information, 0 if not there is none in the picture, and N/A if it should not belong there.
Trsnt_path: Transit shelters must be directly connected to the exterior path of travel and should not obstruct or impede access for pedestrians or mobility device users. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if no transit shelter is present.
Furn_seat: Public benches should be provided, featuring armrests and backrests. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if no seating is present.
Furn_space: There must be a clear space next to the bench or seat for companion seating, ensuring those using mobility devices can sit alongside their companions. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if no seating is present.
Furn_trgt: Tree grates should be level, slip-resistant, and prevent tripping. Landscaping must not overgrow into the accessible path. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if there are no tree grates.
Furn_pr_cd: All protruding objects (e.g., poles, signs) should be above normal persons' height. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if there are no protruding objects in the picture.
Furn_pr_pp: All protruding objects (e.g., poles, signs) should be placed outside the pedestrian path. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if there are no protruding objects in the picture.
Const_rout: When construction affects an accessible route, accessible detours must be clearly marked and navigable. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if there is no construction in the picture.
Ent_wide: Building entrances should be wide enough to accommodate mobility devices and other mobility devices. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if there are no entrances.
Ent_level: Building entrances should be level with the exterior path. If it is elevated, a ramp must be provided, along with steps or stairs. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if there are no entrances.
Ent_obstr: Building doors have no obstructions to prevent them from fully opening. Give a score of 1 if satisfied, 0 if not satisfied, and N/A if there are no entrances.

Pilot and Accuracy

To validate my research, I decided to conduct human tests, which was something I considered within the next steps of the interim report. Thus, I manually audited 500 images, grading the presence and conditions for features concerning my research like sidewalks, transit stops, and building entrances.

Although this may be true, it is important to recognize that humans are not perfect, and conducting human tests comes with subjectivity and mishaps. In relation to the final prompt, it attempts to prevent the problems of subjectivity by providing a more direct, unbiased answer that is strictly based on what is detected and not detected. The grade scores are in direct correlation to the prompt, which is as follows:

If the image contains the feature that meets the criteria, give a score of 1
If the image contains the feature but it does not fully meet the stated criteria, give a score of 0
If the image does not contain the feature, give a score of null (N/A)
If the image contains the feature, but you cannot determine if it meets the criteria due to obstructions, poor image quality, or other visibility issues, give a score of 99

Percentage Agreement Analysis: Comparing Human Tests with AI Outputs

Percentage agreement refers to the proportion of times two raters are in agreement. In my case, this involves comparing the results of human evaluations with AI-generated outputs. This metric is straightforward to compute and clearly measures alignment between raters.

$$\%Agreement = \left(\frac{Number\ of\ Agreements}{Total\ Number\ of\ Observations}\right) x\ 100$$

Where:
Number of Agreements is the count of instances where both raters (e.g., human and AI) provided the same evaluation.
Total Number of Observations is the total number of evaluations or comparisons made (500 images)

For example, if there were 100 observations and 80 instances of agreement, the percentage agreement would be 80%.

When considering the presence or absence of features, ChatGPT's outputs demonstrated high accuracy compared to human assessments, which is promising. However, there were exceptions, notably with Tactile Walking Surface Indicators (TWSI), building entrances, and seating.

Challenges with Specific Features:

TWSI: ChatGPT struggled with detecting TWSI due to Toronto's unique standard of using brown-colored indicators, whereas most other cities use yellow. This divergence complicates recognition.

Building Entrances and Seating: These features are often located away from main focal points in randomly selected images, making them inherently harder to detect. Even for human raters, identifying and evaluating these features proved challenging.

Percentage Agreement Analysis: Comparing Human Tests with AI Outputs

% Agreement on Presence/Absence

Sdwk_width	Sdwk_2_dev	Sdwk_stable	Sdwk_firm	TWSI_curbc	TWSI_con	TWSI_bumps
97.0	92.8	97.3	97.3	51.0	63.6	67.8
TWSI_dir	Cross_curb	Cross_con	Cross_pshb	Trnst_brdg	Trnst_wea	Trnst_rain
73.0	75.5	80.2	66.3	83.2	87.6	88.9
Trnst_wind	Trnst_seat	Trnst_spc	Trnst_vis	Trnst_sign	Trsnt_path	Furn_seat
89.4	89.9	88.6	88.6	82.2	86.6	88.9
Furn_space	Furn_trgt	Furn_pr_cd	Furn_pr_pp	Const_rout	Ent_wide	Ent_level
90.3	72.5	54.5	56.7	88.9	66.3	67.3
Ent_obstr						
67.6						

The percentage agreement understandably decreased when expanding the analysis to include not just the presence or absence of features but also their conditions (quality). This decline was anticipated due to the added complexity of evaluating conditions. Furthermore, it was expected that the percentage agreement would decrease when grading the quality of features.

Despite this, ChatGPT's performance remained relatively strong in assessing transit-related features such as seating and wind conditions, with only minor dips in alignment with human evaluations. This consistency highlights the AI's reliability in handling more nuanced aspects of feature analysis.

% Agreement on Presence/Absence + Conditions

Sdwk_width	Sdwk_2_dev	Sdwk_stable	Sdwk_firm	TWSI_curbc	TWSI_con	TWSI_bumps
59.7	41.3	27.5	27.7	50.5	62.1	67.3
TWSI_dir	Cross_curb	Cross_con	Cross_pshb	Trnst_brdg	Trnst_wea	Trnst_rain
71.0	62.6	72.0	60.1	80.9	87.4	88.4
Trnst_wind	Trnst_seat	Trnst_spc	Trnst_vis	Trnst_sign	Trsnt_path	Furn_seat
85.9	86.9	85.9	86.9	79.7	84.4	87.6
Furn_space	Furn_trgt	Furn_pr_cd	Furn_pr_pp	Const_rout	Ent_wide	Ent_level
87.9	68.6	38.9	40.3	87.1	54.7	55.2
Ent_obstr						
55.9						

Percentage Agreement Analysis: Comparing Human Tests with AI Outputs

Limitations of Using Solely Percentage Agreement

1. Does Not Account for Agreement by Chance
Percentage agreement does not distinguish between genuine agreement and agreement that occurs randomly. For example, having "present" and "absent" categories, some agreement is inevitable purely by chance, even if the raters are not truly aligned.

2. Prone to Imbalance
When one category is much more frequent in reality than another (e.g., "present" 90% of the time and "absent" 10% of the time), percentage agreement can appear deceptively high because raters are more likely to agree on the dominant category, even if their overall performance is poor. Example in my case, sidewalk features.

Regardless of the limitations of solely applying percentage agreement, the straightforwardness is the biggest strength which is my rationale throughout the pilot stages of my research.

Kappa Statistics

To scale away from the limitations of using solely percentage agreement to validate results, I decided to look into the Kappa Statistic. The Kappa Statistic was used in a previous study by Griew et al. in 2013, which used a photo-based analysis using Google Street View Images in England. In their study, Griew et al. demonstrated that while percent agreement can provide a basic understanding of how often raters align, it fails to account for agreements that may occur purely by chance, particularly in datasets with imbalanced categories. By contrast, the Kappa Statistic adjusts for these chance agreements, offering a more accurate assessment of raters (human and AI). The main takeaway from this is that the kappa statistic accounts for the possibility of agreement occurring randomly. Using the kappa statistic instead of simple percent agreement to validate results in an AI photo analysis (where human and AI are the two raters) provides a more robust measure of reliability, especially in contexts where random agreement is possible (Griew et al., 2013a).

According to Griew et al. (2013b), Kappa is widely recognized in reliability studies and has defined interpretation thresholds:

0.01 - 0.20: Slight agreement
0.21 - 0.40: Fair agreement
0.41 - 0.60: Moderate agreement
0.61 - 0.80: Substantial agreement
0.81 - 1.00: Almost perfect agreement

Kappa Statistics

Kappa Statistic on Presence/Absence + Conditions

sdwk_width	Sdwk_2_dev	Sdwk_stable	Sdwk_firm	TWSI_curbc	TWSI_con	TWSI_bumps
0.129	0.048	0.026	0.03	0.082	0.084	0.119
TWSI_dir	Cross_curb	Cross_con	Cross_pshb	Trnst_brdg	Trnst_wea	Trnst_rain
0.066	0.414	0.562	0.198	0.281	0.355	0.369
Trnst_wind	Trnst_eat	Trnst_spc	Trnst_vis	Trnst_sign	Trnst_path	Furn_seat
0.248	0.281	0.302	0.343	0.283	0.295	0.288
Furn_space	Furn_trgt	Furn_pr_cd	Furn_pr_pp	Const_rout	Ent_wide	Ent_level
0.25	0.082	0.069	0.069	0.263	0.262	0.254
End_obstr						
0.254						

- Features with low kappa scores (e.g., Sidewalk Stability, Sidewalk Width for 2 Mobility Devices) indicate that these aspects are harder to evaluate consistently due to subjective interpretation or lack of clear criteria.
- Features with moderate kappa scores (e.g., Pedestrian Crosswalk Curb Ramps, Transit Seating) show a moderate agreement, suggesting that these characteristics are easier to evaluate consistently.

Going forward, it will be important to focus on improving the reliability of features with low kappa scores by refining the evaluation criteria (prompt), for both human and AI raters. Moreover, features with lower kappa values could benefit from more specific definitions and more precise rating guidelines. In comparison, features with higher kappa values suggest that the current assessment methods are heading in the right direction.

BIA Selection

The Crossroads of the Danforth BIA was selected for its vibrant community and the unique challenges faced by its residents. Demographic analysis revealed significant resource gaps and accessibility features that hinder equitable inclusion. A notable challenge is linguistic accessibility, with 46% of the population speaking a language other than English and 14.6% having limited English proficiency. Despite the area's rich cultural diversity—63.8% of residents identify as visible minorities—there is a lack of multilingual signage and culturally sensitive services, making it difficult for non-English-speaking residents to access businesses and community resources.

Economic disparities further emphasize the need for intervention. The median household income in the area is $57,000, significantly lower than the Toronto average of $97,000 (CHMC, 2021). Nearly half of households spend more than 30% of their income on housing, and renters are disproportionately affected by unaffordable shelter costs. Seniors, who comprise 16% of the population and often live alone, face compounded challenges due to limited mobility and affordability. The area's reliance on public transit (used by 49% of commuters) also highlights accessibility issues. Long commute times, coupled with insufficient physical infrastructure such as narrow sidewalks, limited ramps, and inaccessible entrances, create barriers for economically disadvantaged and working-age populations.

Feature Selection

To address these challenges, I chose to map specific features that directly impact mobility and safety. These include crosswalk contrast, sidewalk width (for one and two mobility devices), accessible seating within transit shelters, and the presence of Tactile Walking Surface Indicators (TWSI) on curb cuts. These features were prioritized because they are critical for ensuring inclusivity and ease of movement in public spaces. Mapping these features allows for a spatial understanding of current conditions and highlights areas for improvement, providing actionable insights to enhance accessibility.

	0s	1s	99s	N/A
Sdwk_width	311	317	71	45
Sdwk_2_dev	483	0	70	191
Sdwk_stable	37	43	621	43
Sdwk_firm	37	43	621	43
Sdwk_drain	14	62	344	324
TWSI_curbc	0	9	1	734
TWSI_con	3	5	2	734
TWSI_bumps	0	9	1	734
TWSI_dir	1	0	6	737
Cross_curb	90	218	39	397
Cross_con	111	175	12	446
Cross_pshb	9	32	10	693
Trnst_brdg	11	28	14	691
Trnst_wea	1	24	5	714
Trnst_rain	1	24	5	714
Trnst_wind	22	1	6	715
Trnst_seat	22	1	6	715
Trnst_spc	22	6	11	705
Trnst_vis	1	28	9	706

My selection of features was also driven by data. High occurrences of (0) values—indicating poor-quality features not meeting accessibility standards—and null values, where features were entirely absent, underscore the urgency of addressing these gaps. By focusing on these quantifiable and impactful elements, the project lays a foundation for meaningful improvements that benefit all residents, especially those with mobility challenges.

Mapping Accessibility Conditions

These selections focus on mapping and visualizing the current accessibility conditions within The Crossroad of the Danforth Business Improvement Area (BIA) to identify critical gaps and challenges. By analyzing specific features such as sidewalk widths, tactile walking surface indicators (TWSI), transit seating, and crosswalk contrast, I have created a detailed visual representation of accessibility in the area. The data highlights various issues, including areas where sidewalks are too narrow for mobility devices, where pedestrian crossings lack proper markings, and where tactile indicators on curb cuts are missing. Additionally, gaps in accessible transit seating, and inconsistencies in sidewalk design for accommodating multiple mobility devices. These maps not only reveal the locations of accessibility gaps but also emphasize systemic challenges, such as incomplete infrastructure, obstructions, and a lack of proper design considerations for inclusive mobility. This visual analysis aims to provide a clear understanding of the accessibility landscape in the BIA and serve as a foundation for addressing these gaps to more equitable urban environments.

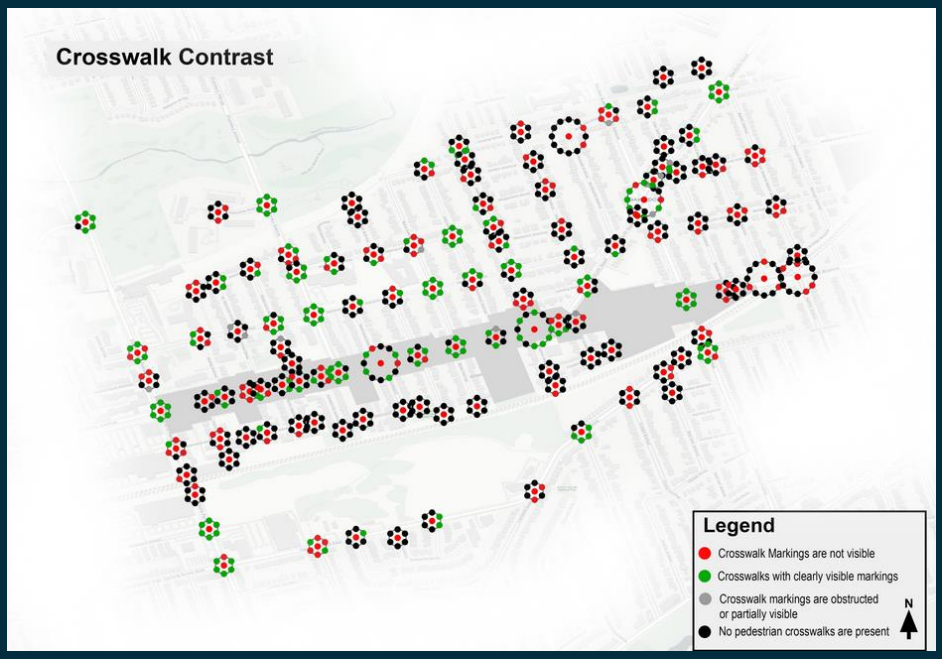

Figure 1.0:
Explanation of Crosswalk Contrast Map: This map visualizes the presence and quality of crosswalk across the BIA.
Red Points (0 Values): Areas lacking sufficient crosswalk contrast, creating potential safety hazards for pedestrians, especially those with visual impairments.
Green Points (1 Values): Locations where crosswalks meet the required contrast for visibility, ensuring safer pedestrian crossings.
Gray Points (99 Values): Crosswalks where obstructions or poor image quality made it impossible to assess the condition.
Black Points (Null Values): Locations where crosswalks are not present highlighting potential gaps in pedestrian infrastructure.

Please see appendix for more maps

Recommendations

Unitil now, this report has focused on the methods and visualization of the data gathered from the processes. Applying this data to communities and built environments will be the focus of this section. It will begin with general considerations when making recommendations using this data and will then be followed with an example of a recommendation report given the data I obtained regarding accessibility features in the Crossroads of the Danforth BIA.

It is important first to recognize that this technology must be seen as one tool in a toolkit for auditing and assessing public spaces and built environments. It should not be used in isolation to direct resources or make complete recommendations. The use cases for this technology could be to provide macro-level assessments and direct attention to certain areas where accessibility features are lacking. It could be useful to search for specific features and the prevalence of their presence across large areas. The Accessibility for Ontarians with Disabilities Act has been in force since 2005, yet still remains toothless and lacks enforcement. The critical Fourth Review of the AODA completed by Rick Donovan states that a key obstacle is the lack of human resources for the 460,000 organizations subject to the act. This technology could provide efficient, cost-effective assistance to this matter. (ARCH Disability Law Centre, 2023)

It is also vitally important to recognize that in the hierarchy of feedback, people with lived experience should be consulted and heard above this technology especially with how accessibility features are implemented. As Matthew Shaw from the Rick Hansen Foundation stated in advice to the mee, accessibility is a state of being and not a checklist of what features are or are not there. This mindset calls for people to be able to access spaces for its intended purpose and in doing so, provides a more equitable community opening the doors to more opportunities. It is against this backdrop that these recommended considerations are suggested.

Recommendations

To help organizations, municipalities, and other stakeholders understand the importance of the findings, it is vital for them to understand the motivation and rationale to implement the suggestions. Ontario is an aging society where 40% of Ontario's population is over 65 suggesting that accessibility will likely touch most of their lives in the very near future. (Legislation, 2024) As such, an emphasis on aging in place has come to the fore and would require neighbourhoods, BIAs, and public spaces to have more rest areas, ramps, and curb cuts. Aside from the aging population, any able-bodied person is susceptible to accidents that could result in reduced mobility or impairments and would benefit from an accessible city.

Inclusivity and equity are values that ought to be goals for any society that values human dignity. The social model of disability argues that people are not disabled by their impairments, but the disabling barriers that people face in their society. (Oliver, 2013) This relocates disability from an internal perspective that prevailed in the medical model before it gave way to the social model. This understanding implies that society is structured to disable people's participation in it. The findings from this technology then, can be a way to identify and deconstruct the barriers to disability.

With this in mind, removing barriers and increasing the potential for people with disabilities to participate in all aspects of society could allow for a broader labour force and economic contributions. A 2022 report on Canadian labour force participation rates for people with disabilities stated that able-bodied people aged 16-65 had an employment rate of 80.1% compared to the 65.1% for people with disabilities. This was also further broken down by disabilities ranging from mild, moderate, severe, and very severe with employment rates of 76.6%, 64.5%, 50.4%, and 26.8% respectively. The report also noted that people with disabilities often worked part-time jobs, were paid less, and women with disabilities had higher rates of unemployment than men with disabilities. (Canada, 2022) Those living on Ontario Disability Support Program (ODSP) can only receive a maximum of $1,169 per month and should they find a job, will see their ODSP reduced after reaching a certain threshold of income which, in effect, discourages part time work at low wages even though this is the most common type of employment for people with disabilities.(Saba, 2022) Providing gainful employment with fair wages could greatly increase a person's life and this begins with an equitable society and public realm.

The economic impact goes hand in hand with the employment aspect where people with disabilities who are able to have a sufficient income need access to the businesses to spend it. It is not only people with disabilities that would like to access small, local businesses, but the aging adults as well. This is a vital consideration for municipalities and, especially, BIAs where storefronts may not have accessible entrances.

Considerations

→ Community Participation

This is a novel audit tool in its infancy and one that does not capture the nuances in a streetscape. Nothing here should be considered definitive especially without the input from people with lived experience. It is their participation in audits that would provide precise and critical information in implementing any accessibility changes. Matthew Shaw (2024) has told us of the mantra: Nothing for us without us.

→ Funding

Grants are an integral part to implementing change as accessibility features are often poorly funded. Awareness of grants and funding sources can help prioritize which features can be implemented immediately and/or what can be the most cost effective. Unfortunately, these grants can be difficult to find making this aspect of the recommendations a useful resource to provide to the audience. The City of Toronto's underutilized Facade Improvement Grant Program provides funding for commercial property owners to enhance building facades, including accessibility features like ramps and automatic doors. (Macrae, 2024)

→ Policy

Common issues across the city should be combined and brought to the local government's attention to reform policy. Doing so would provide consistent features across the city, an important aspect of accessibility that people with disabilities rely on. Being able to interact in the same way with accessibility features allows people with disabilities to navigate the city safely and predictably. Policy should also be reviewed regularly to ensure that consistency is maintained. This will provide accessibility features that are up to standard and evenly distributed throughout the city.

→ Timelines

A three tier timeline should be considered when evaluating how to allocate funds equitably throughout the city such as short term, medium term and long term. A short term timeline would be defined as a project that requires little funding and can be easily fixed or repaired. This will allow for repairs to be done quickly and function to its full capacity. A medium term timeline refers to accessibility design that requires a bit more funding but also could wait a couple of years before it goes through repair. This model demonstrates urgency but not immediate attention and can be planned around the budget for that year. Then the long term model refers to setting aside funds for larger projects or designs that are planned in advance for installment or repair. This model is not perceived as urgent and can be held off for 3-5 years, however the project is still necessary and does require attention. These models help with structuring order for allocating funds for accessibility designs to ensure that all designs are given equal attention and repairment to avoid a city wide repairment budget.

Sample Recommendations

Rationale

The Rationale for the Crossroads on the Danforth BIA would combine the information at the beginning of the Recommendations section and the demographic information in the BIA selection section.

Short Term Interventions (0-1 year)

Community Engagement Campaign

Use interactive feedback methods such as information boards with stickers (green, yellow, red) and QR codes linked to detailed recommendations and surveys. Conduct pop-up sessions at high-traffic areas (transit stops, parks, community centers) to gather feedback on accessibility priorities.

Rationale

This builds momentum for accessibility improvements, directly incorporates community voices, and aligns with the "nothing for us without us" principle.

Funding

Apply for small-scale grants like the "Canada Healthy Communities Initiative" or the "Neighbourhood Support Grant (Toronto)" to fund community consultation materials.

Short Term Interventions (0-1 year)

Sidewalk Improvements

Repair cracks, clear obstructions, and add temporary ramps in high-traffic zones. Use AI or mapping tools to identify priority areas based on gaps identified in test results.

Rationale	Funding
Sidewalk stability (87.7%) is a strength, but inconsistent widths and drainage issues disrupt meaningful access. Targeted fixes create a more continuous experience.	"Ontario Main Street Revitalization Initiative" may fund minor sidewalk upgrades.

Tactile Walking Surface Indicators

Add tactile paving at major intersections and transit stops.

Rationale	Funding
Supports visually impaired residents by improving navigation. This is a small but impactful step toward inclusive design.	None

Accessible Signage

Install temporary multilingual signs with high-contrast colors at key intersections and public spaces. Use universally recognizable symbols for accessibility.

Rationale	Funding
Reflects the demographics in the area.	"Ontario Trillium Foundation (OTF) Seed Grants" for initiatives that build community inclusion.

Medium Term Interventions (1-3 years)

Universal Design Building Audits

Use Rick Hansen Accessibility Certification (RHFAC) or similar evaluation tools to audit key public spaces and businesses for accessibility.

Rationale	Funding
Comprehensive audits ensure all future improvements align with accessibility standards.	Explore "Enabling Accessibility Fund (EAF) – Mid-Sized Projects" for grants up to $1 million to retrofit or build accessible spaces.

Improved Public Seating

Install permanent, universally designed seating along major routes, parks, and waiting areas. Ensure clear spaces for mobility aids and shade for seniors.

Rationale	Funding
The current lack of accessible seating disproportionately impacts seniors and those with mobility impairments.	None

Technology Integration for Data Collection

Deploy AI-based solutions for ongoing accessibility mapping and monitoring. Consider partnerships with universities or tech firms to pilot innovative data collection methods.

Rationale	Funding
Accurate, real-time data informs better decision-making without the exhaustive costs of manual audits.	None

Medium Term Interventions (1-3 years)

Business Accessibility Partnership Program

Launch a program to subsidize accessibility upgrades (e.g., ramps, automatic doors) for local businesses.

Rationale

Businesses often lack funds for upgrades, yet accessible shops are vital for a seamless community experience.

Funding

Use the "Enabling Accessibility Fund (Small Projects)" for grants up to $100,000.

Long Term Interventions (3+ years)

Comprehensive Infrastructure Upgrades

Redesign sidewalks and streetscapes with continuous, wide pathways, curb cuts, and proper drainage. Integrate accessibility features into all new infrastructure projects.

Rationale

These upgrades address core housing and mobility needs, ensuring a seamless experience throughout the area.

Funding

""Infrastructure Canada's Investing in Canada Plan" for large-scale urban improvements and the City of Toronto's City Facade Improvement Grant Program for commercial frontage improvements including accessibility features like ramps and automatic door openers.

Long Term Interventions (3+ years)

Policy Review

Review other BIAs to determine whether or not they are experiencing similar accessibility design and implementation issues

Rationale	Funding
Provide consistent accessibility standards across the city ensuring that everything is uniform and no one is left behind	None

Community Accessibility Hubs

Develop multi-use spaces offering accessible services such as language classes, mobility equipment rentals, and health resources.

Rationale	Funding
Provides ongoing support for seniors, low-income residents, and immigrants while promoting inclusivity.	"Ontario Community Infrastructure Fund (OCIF)" can support larger-scale community projects.

Redesign Transit Shelters

Work with the Toronto Transit Commission (TTC) to create new standards for transit shelters and stops to allow for more rest areas and clear ground area

Rationale	Funding
These changes allow for safer shelter from harsh weather and long wait times	None

Conclusion

By strategically layering short-term fixes, medium-term projects, and long-term infrastructure commitments, the Crossroads of the Danforth BIA can achieve meaningful, cumulative access improvements. Leveraging grants such as the Enabling Accessibility Fund and the Canada Healthy Communities Initiative ensures financial feasibility while engaging the community guarantees that solutions remain inclusive and user-centered. Together, these recommendations align with the vision of accessibility as an ongoing commitment to serve all users.

This report demonstrates the potential ability of ChatGPT for evaluating accessibility in today's urban environments. The AI alongside my prompt was able to identify many key accessibility features, and while it wasn't perfect, it still greatly improved over my testing. Key challenges like the source of my images, as well as the AI's greater understanding of accessibility will need to be addressed before any scaled adoption of this technology. Regardless, ChatGPT and other image reading AIs show promise in not only evaluating accessibility but potentially aligning with urban planning needs at a greater scale. Through this report's findings, the importance of a refined prompt with clear instruction, as well as a continued effort to test accuracy is evident. Ultimately both AIs ability as well as ours will have to advance before AI is widely used to evaluate accessibility.

APPENDIX

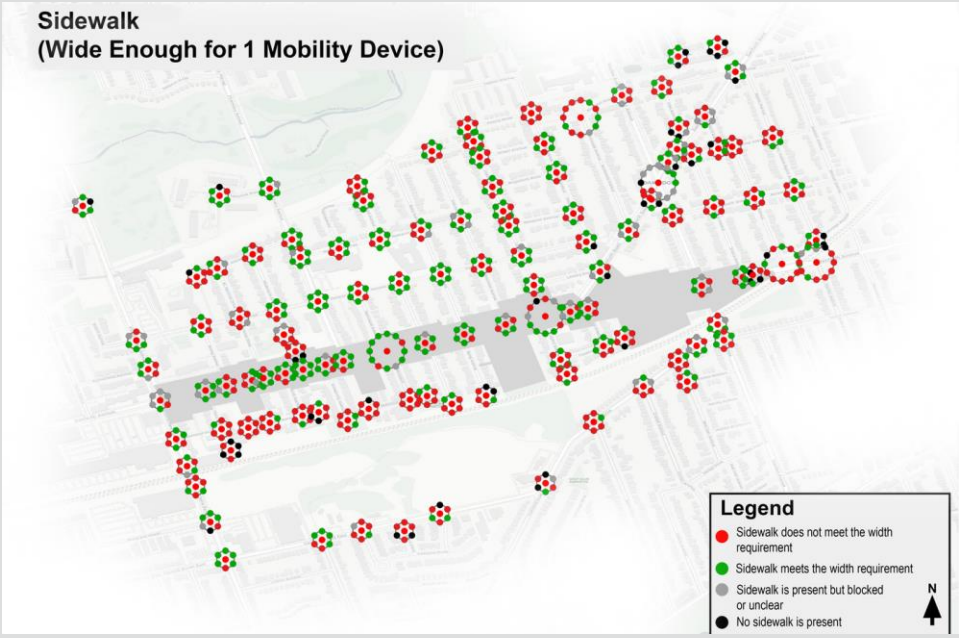

Figure 2.0:
Explanation of Sidewalk Width Map: This map illustrates the adequacy of sidewalks in the BIA for accommodating one mobility device.
Red Points (0 Values): Sidewalks that fail to meet the required width, creating barriers for mobility device users.
Green Points (1 Values): Sidewalks that meet the minimum width requirement for mobility devices, indicating accessible infrastructure.
Gray Points (99 Values): Areas where obstructions or poor image quality prevent accurate assessment.
Black Points (Null Values): Locations without sidewalks, highlighting missing pedestrian pathways.

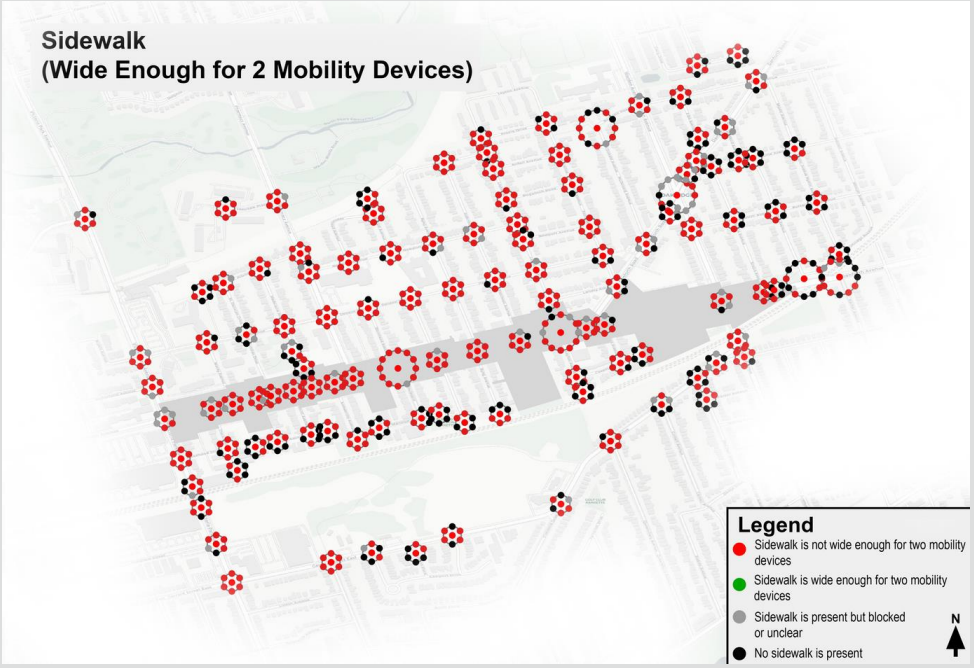

Figure 3.0:
Explanation of Sidewalk Width for Two Mobility Devices Map: This map evaluates sidewalk widths within the BIA to determine if they can accommodate two mobility devices side by side.
Red Points (0 Values): Sidewalks that fail to meet width requirements, posing challenges for mobility device users.
Green Points (1 Values): Sidewalks that are sufficiently wide for two mobility devices, ensuring accessibility for diverse users.
Gray Points (99 Values): Locations where obstructions or poor image quality hinder assessment.
Black Points (Null Values): Areas where sidewalks are absent, reflecting infrastructure gaps.

Figure 4.0:
Explanation of Accessible Seating within Transit Shelter Map: This map assesses the presence of accessible seating within transit shelters in the BIA.

- Red Points (0 Values): Transit shelters lacking accessible seating, indicating a gap in transit infrastructure.
- Green Points (1 Values): Transit shelters equipped with accessible seating, meeting the needs of mobility device users.
- Gray Points (99 Values): Areas where obstructions or poor image quality prevent accurate evaluation.
- Black Points (Null Values): Locations without transit shelters, highlighting missing facilities.

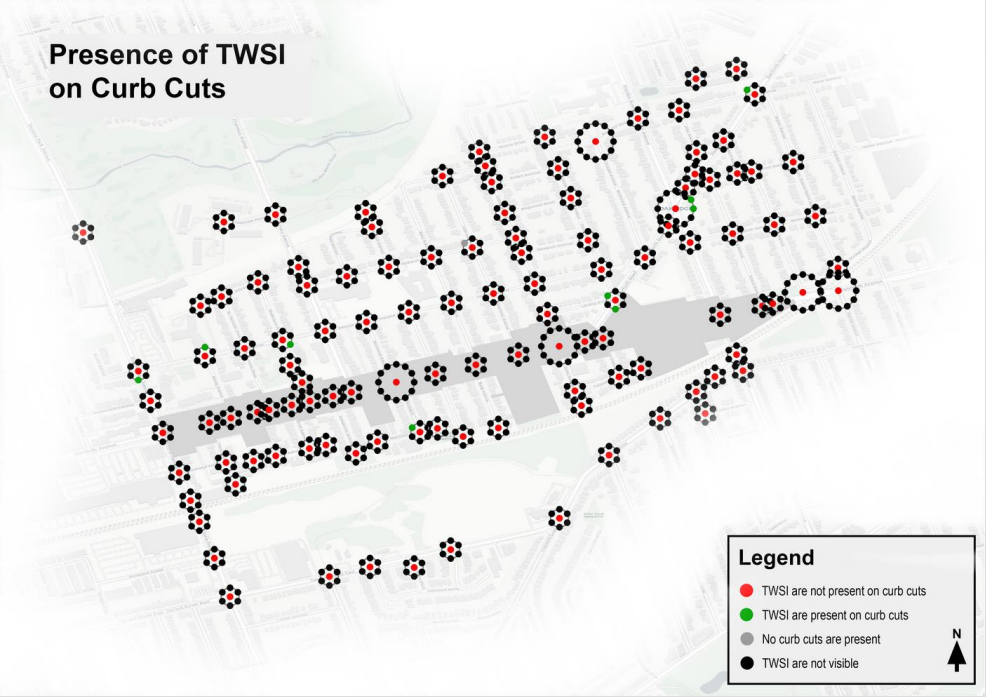

Figure 5.0:
Explanation of TWSI Presence on Curb Cuts Map: This map assesses the presence of Tactile Walking Surface Indicators (TWSI) on curb cuts in the BIA.

- Red Points (0 Values): Curb cuts lacking TWSI, posing safety and accessibility challenges.
- Green Points (1 Values): Curb cuts with TWSI present, ensuring accessibility for visually impaired pedestrians.
- Gray Points (99 Values): Locations where obstructions or image quality hinder evaluation.

Black Points (Null Values): Areas without curb cuts, indicating gaps in pedestrian infrastructure.

Figure 6.0
Sidewalk Accessibility Map Evaluating Curb Cuts: This map evaluates the sidewalks accessibility in regards to curb cuts
Red Points: The red dots indicate points deemed N/A for TWSI/Curb Cuts

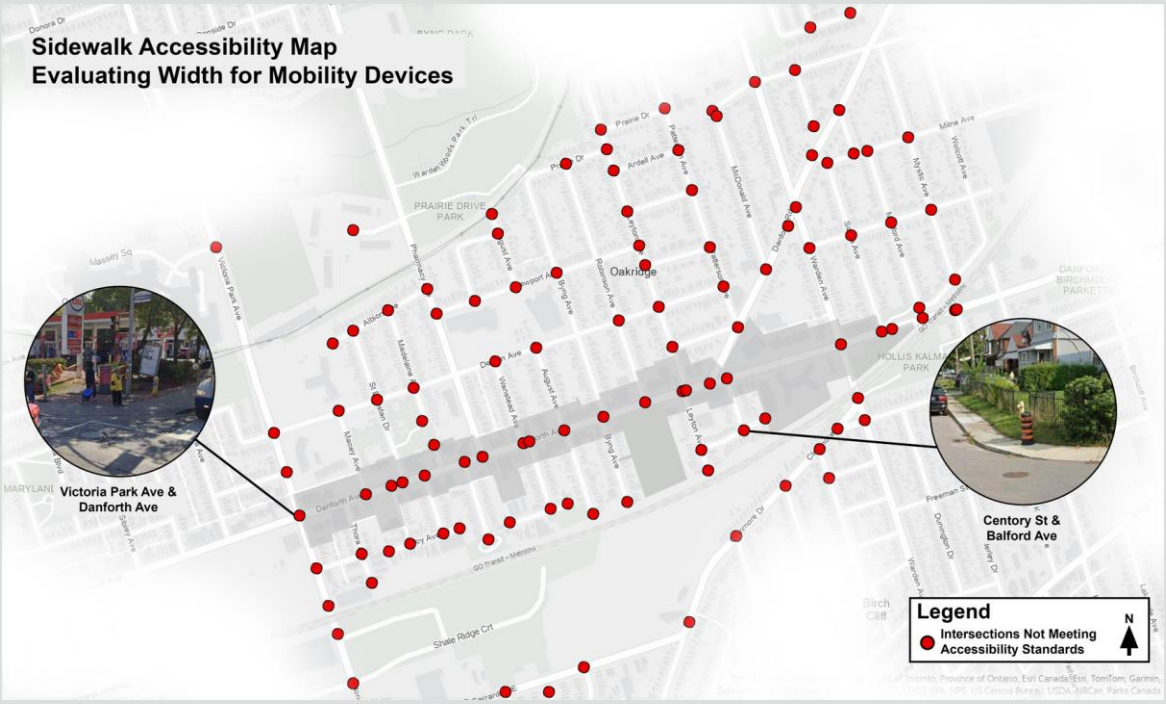

Figure 7.0
Sidewalk Accessibility Map Evaluating Width for Mobility Devices

- Red Points: The red dots indicate points where sidewalks, as detected by ChatGPT, are poor quality and are not wide enough for one mobility device.
- On the bottom right callout image is a sidewalk on the right side of the road. However, there is no sidewalk on the left side. AI has been showing inconsistencies and might only be partially accurate in its detection due to the absence of a sidewalk on one side of the road. This limitation in the AI's analysis could stem from its inability to account for scenarios where sidewalks exist only on one side, leading it to flag areas incorrectly.

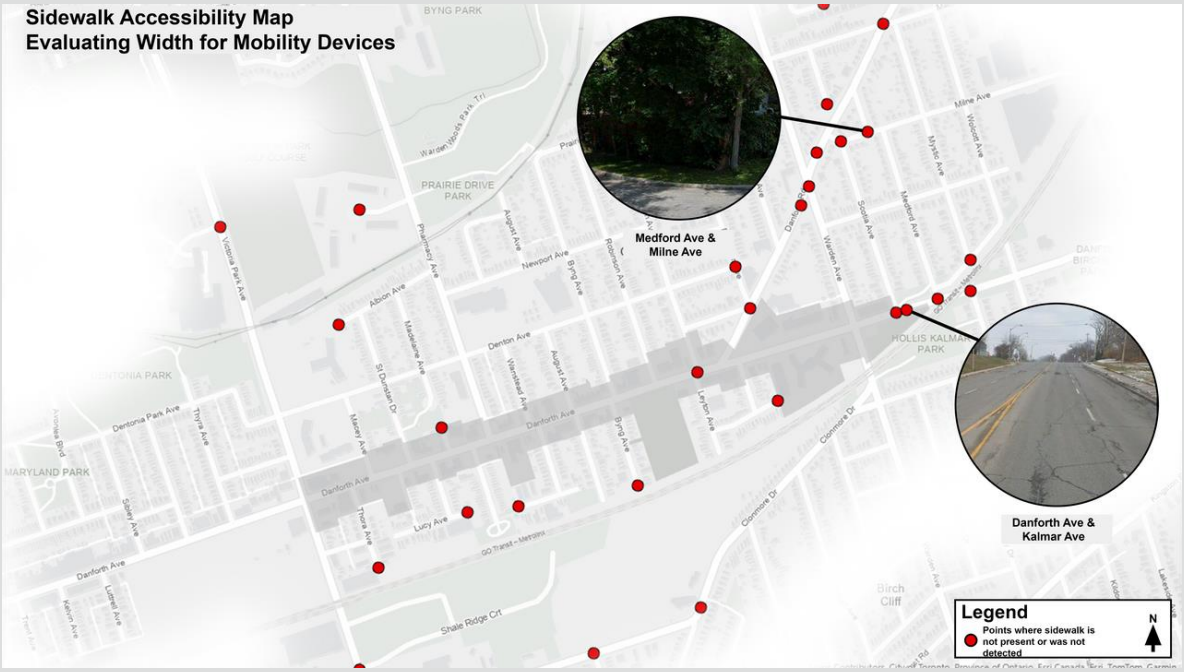

Figure 7.1
Sidewalk Accessibility Map Evaluating Width for Mobility Devices

- Red Points: The red dots indicate points where sidewalk is not present or was present, but not detected by AI
- On the bottom right callout image is a sidewalk on the right side of the road. However, there is no sidewalk on the left side. AI has been showing inconsistencies and might only be partially accurate in its detection due to the absence of a sidewalk on one side of the road. This limitation in the AI's analysis could stem from its inability to account for scenarios where sidewalks exist only on one side, leading it to flag areas incorrectly.

Figure 8.0
Transit Accessibility Map Evaluating Transit Stops For Seating: This map evaluates the transit accessibility in regards to seating for transit stops
Red Points: The red dots indicate points deemed at value 0 for the Transit stop setting for not meeting standards.

Figure 8.1
Transit Accessibility Map Evaluating Transit Stops for Seating
Red Points: The Red dots indicate points where transit stop seating could not be determined

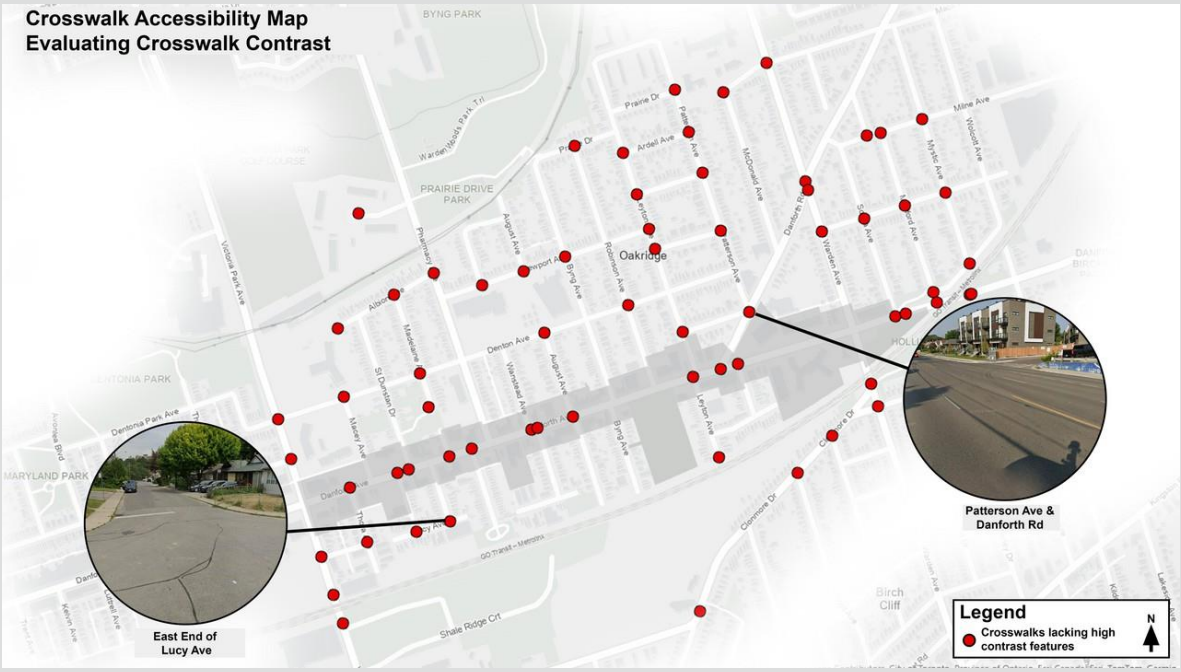

Figure 9.0
Crosswalk Accessibility Map Evaluating Crosswalk Contrast

- Red Points: The red dots indicate points with no crosswalks
- On the top right image callout, a pedestrian crosswalk is present but was not detected.

Figure 9.1
Crosswalk Accessibility Map Evaluating Crosswalk Contrast
Red Points: The red dots indicate points where crosswalks are lacking high-contrast features, failing accessibility standards

References

Accuracy Validation

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Recommendations

1. ARCH Disability Law Centre | Submission to Rich Donovan – 4th Independent Review of the Accessibility for Ontarians with Disabilities Act. (2023). Archdisabilitylaw.ca. https://archdisabilitylaw.ca/submission-to-rich-donovan-4th-independent-review-of-the-accessibility-for-ontarians-with-disabilities-act/
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